Codefest 1
Hardware for Artificial Intelligence and Machine Learning (HW4AI)
ECE 410/510
Spring 2026

Overview

This codefest has two sequential challenges. Complete CMAN before opening CLLM. CMAN must be completed without any AI assistance; close all LLM tabs before starting. CLLM permits and often requires AI tooling. Topic: What is computation? AI/ML landscape. Deliverables are committed to your GitHub repository by Sun, Apr 5, 11:59 pm. Each codefest is graded *Satisfactory (S) / Not Yet (NY)* as a CMAN+CLLM pair. A late submission counts as *Not Yet* and consumes one revision token.

CMAN — Workload accounting by hand

You are given a 3-layer fully connected network with dimensions [784 → 256 → 128 → 10], batch size 1, all weights and activations in FP32 (4 bytes each). No bias terms. Complete the following tasks by hand, without using a LLM.

Tasks

1. For each layer, compute the number of multiply-accumulate operations (MACs). Show the formula and the substituted values.
2. Sum the MACs across all three layers to get the total MACs for one forward pass.
3. Compute the total number of trainable parameters (weights only, no biases).
4. Compute the total weight memory in bytes (FP32).
5. Compute the total activation memory in bytes needed to store the input and all layer outputs simultaneously (FP32).
6. Compute arithmetic intensity as: (2 × total MACs) / (weight bytes + activation bytes).

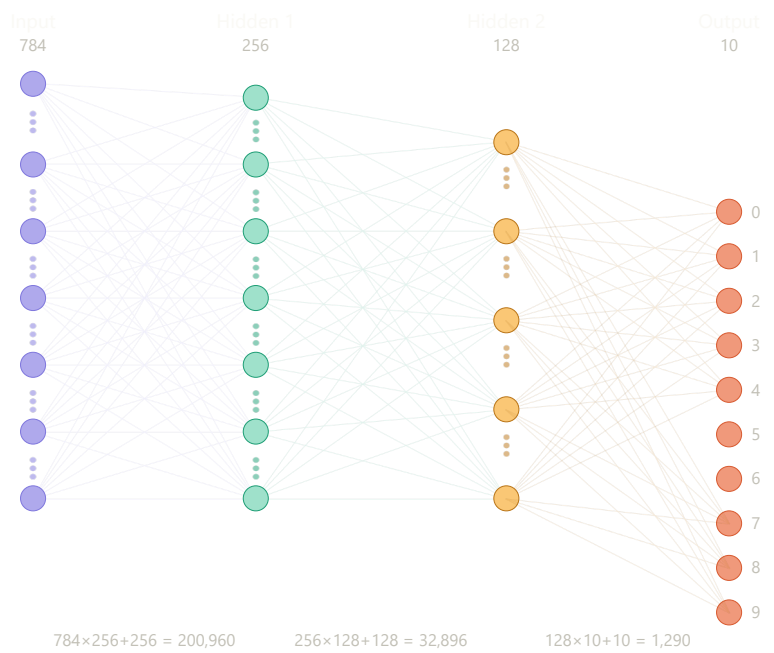

Figure 1: A 3-layer fully connected network with dimensions [784 → 256 → 128 → 10].

CMAN NO AI	Deliverable:
	• one file (handwritten photo as PDF/PNG, or .md). Must contain:
	  ○ (a) per-layer MAC table with formula and values shown;
	  ○ (b) total MACs;
	  ○ (c) total parameters;
	  ○ (d) weight memory in bytes;
	  ○ (e) activation memory in bytes;
	  ○ (f) arithmetic intensity in FLOP/byte
	  ○ File: `codefest/cf01/cman_workload_accounting.md` (or .pdf/.png)

CLLM — Environment setup and ResNet-18 profiling

Set up your working environment and profile a real model to practice the same analysis at scale, by using tools, including LLM. Commits are to the GitHub repository you created before this codefest. If you have not shared your Github URL yet, please do so in the spreadsheet linked on Canvas.

ResNet-18 is a Convolutional Neural Network (CNN) architecture with 18 layers. ResNet-18 is widely used for computer vision tasks like image classification. `torchinfo` prints a detailed summary table of the model architecture: layer names, output shapes, parameter counts per layer, and estimated memory usage. It doesn't execute the forward pass; it infers dimensions from the architecture. `torchinfo` is good for inspecting model structure and layer-wise parameter distribution.

Tasks

1. If you haven't done so yet, create your course GitHub repository. The repo must be public. Add a README.md containing your name, the course (ECE 410/510 Spring 2026), and your tentative project topic candidate. Push the initial commit before the end of the session.
2. Install `torchinfo` (`pip install torchinfo`). Profile ResNet-18 on a single FP32 forward pass (batch=1, input 3×224×224). Save the full `torchinfo` output as `codefest/cf01/profiling/resnet18_profile.txt`. If necessary, ask your LLM on how to use `torchinfo` to profile ResNet-18.
3. Identify the five layers with the highest MAC count. Record their names, MACs, and parameter counts in a markdown table in `codefest/cf01/profiling/resnet18_analysis.md`. Markdown table:
   - https://www.codecademy.com/resources/docs/markdown/tables
   - https://www.tablesgenerator.com/markdown_tables
4. For the single most MAC-intensive layer, compute its arithmetic intensity assuming all weights and activations are loaded from DRAM with no reuse. Show the calculation in `codefest/cf01/profiling/resnet18_analysis.md`.

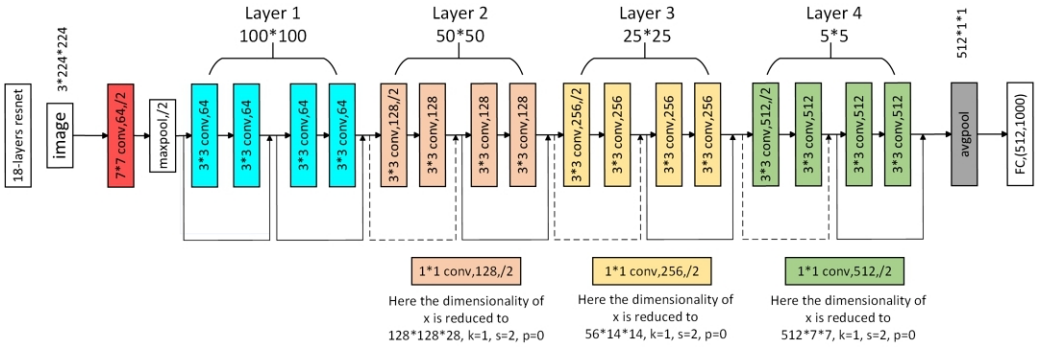

Figure 2. Structure of ResNet-18. Source: https://dl.acm.org/doi/fullHtml/10.1145/3613917.3613922

CLLM AI OK	Deliverable: • GitHub repository containing: • `codefest/cf01/cman_workload_accounting.md` (or .pdf/.png) • `codefest/cf01/profiling/resnet18_profile.txt`

	• `codefest/cf01/profiling/resnet18_analysis.md` (top-5 MAC table + AI calculation) • `project/heilmeier_draft.md` (first 3 questions answered) • `project/algorithm_diagram.png` (or .pdf) • Repo must be public and accessible by Sunday 11:59 pm.

Project — Heilmeier questions

1. Draft answers to the first 3 Heilmeier questions (https://www.darpa.mil/about/heilmeier-catechism) for your project idea. Save as `project/heilmeier_draft.md`.
2. Draw a high-level block diagram of your target algorithm (hand-drawn photo or use any tool, e.g., your LLM or PyFlowchart).
3. Save as `project/algorithm_diagram.png` (or .pdf).